

Eldon Dahlin

Software Engineer | Agentic AI, LLM Applications & RAG Pipelines in Python

eldondahlin@gmail.com · (505) 385-8009 · Denver, CO - Open to relocation (SF / NYC) · GitHub: github.com/end1989 · LinkedIn: linkedin.com/in/eldon-dahlin

Full-stack Software Engineer with 7+ years building backend services, large-scale data pipelines, and AI-integrated tooling in Python, TypeScript, and Java. Hands-on with LLM-powered applications, RAG pipelines, embeddings, semantic search, and MCP-based AI-assistant integration. Builds AI-safety tools on my own time - including VerifyMyAI, an open-source system detecting prompt-injection and jailbreak instructions - with a focus on responsible, secure, and reliable AI.

Experience

Software Engineer *Travelport*

2021-07–Present · Denver, CO (Remote)

- Built a proof-of-concept MCP (Model Context Protocol) server enabling AI-powered querying over platform data - internal tooling that accelerated investigation
- Developed an automated data-quality assessment that benchmarks provider accuracy against an independent source of truth (Google Places API), exposing discrepancies at scale and quantifying which providers could be trusted
- Built validation and monitoring systems that detect bad and anomalous records across a hotel-content platform of 5.5M+ entries from global providers (Booking.com, GIATA), protecting every downstream system that consumes the data
- Engineered ingestion pipelines processing ~4M Booking.com and ~1.5M GIATA records (static data and imagery) across multiple provider feeds
- Proposed and designed an end-to-end data-protection and cleanup framework to catch corruption before it propagates across the pipeline
- Developed a CHD validation library in Java / Spring Boot for enterprise-grade data validation across the pipeline
- Designed an accommodation-type classification system to standardize categorization across disparate, messy provider data sources

Software Engineer *Dish Network*

2018-12–2021-07 · Denver, CO

- Designed and architected high-traffic REST APIs serving customer account and service data on a small, collaborative team
- Built Java microservices for customer-data management with Spring Boot, Kafka event streaming, and Docker
- Protected data integrity by detecting and correcting corrupted encrypted customer records before they could cause downstream loss
- Wrote Python automation to streamline data operations and reduce manual intervention
- Drove adoption of stronger Agile practices and team conventions

Projects

VerifyMyAI - Open-Source AI-Safety Tool *React 19, Vite, Tailwind*

verifymyai.org

- Built a free, open-source tool that detects hidden, coercive instructions (prompt-injection / jailbreak) buried in consumer AI assistants (ChatGPT, Claude, Gemini, Copilot)
- Engineered a detection-prompt library (12 targeted audits) and a red-flag pattern set covering six manipulation patterns
- Designed a fully client-side experience with zero tracking, accounts, or data collection - built around responsible AI and data-privacy principles

Hybrid RAG Search Engine *Python, ChromaDB, Sentence Transformers, MCP*

- Built a hybrid semantic + keyword retrieval system (BM25 + dense embeddings) with an MCP server interface for AI-assistant integration and multi-profile knowledge bases

AI Content-Classification Pipeline *Python, PyTorch, CLIP, FastAPI*

- Built an end-to-end ML pipeline using CLIP for zero-shot image classification and PaddleOCR for text extraction, with active learning for continuous improvement
- Included automated content-moderation classification and GPU-accelerated (CUDA) batch inference behind a FastAPI backend

Task Orchestration Engine *Python, FastAPI, Docker*

- Architected a manager-worker distributed system with process isolation, resource limits, and automated review-and-approval gates
- Implemented task queuing with status transitions, secret-detection scanning, artifact management, and full audit-trail logging

Skills

Languages: Python, TypeScript, JavaScript, Java (8/17/21), SQL, Bash

AI / LLM: LLM integration (Claude / Anthropic API & local models), prompt engineering, RAG pipelines, MCP (Model Context Protocol) servers, embeddings, semantic search, classifiers, CLIP, PyTorch, Sentence Transformers, scikit-learn, prompt-injection / jailbreak detection (VerifyMyAI)

Backend: FastAPI, Flask, Spring Boot, Express / Node.js, REST APIs, Server-Sent Events

Data: ChromaDB (vector search), MongoDB (aggregation pipelines, large-scale ingestion), PostgreSQL, Oracle, SQLite

Infra & Tooling: Docker, Docker Compose, Kafka, Git, GitHub Actions, GitLab CI, Jenkins

Detection & Integrity: Abuse, misuse & anomaly detection, Trust & Safety / content moderation, data-quality & integrity engineering, validation & policy-enforcement frameworks, monitoring, audit logging

Practices: Microservices, event-driven architecture, CI/CD, Agile / Scrum, SOLID, data governance

Education

Full-Stack Web Development Immersive *Galvanize*

2018

Web Development *Sacramento City College*

2017 - 2018

Additional

- Languages: English (native), Spanish (conversational - lived in Peru 2008-2010)
- Open source: github.com/end1989